① CH#01 Complex Nos.

$-\pi < \theta \leq \pi$ ↳ Principal Argument of z

$z = x+iy \Rightarrow \bar{z} = x-iy,\ |z| = \sqrt{x^2+y^2}$

$\Downarrow$

$z = r\cos\theta + ir\sin\theta = r(\cos\theta + i\sin\theta),\ r = \sqrt{x^2+y^2},\ \theta = \tan^{-1}\left(\frac{y}{x}\right)$

$\pi-\theta$	θ
$\theta-\pi$	$-\theta$

$-\pi < Arg(z) \leq \pi$, $e^{i\theta} = cis\,\theta = \cos\theta + i\sin\theta$.

↳ Euler's Formula

Additive Identity of Complex No's $= (0,0)$

Multiplicative Identity of Complex Nos. $(1,0)$

$Arg(z_1z_2) = Arg(z_1) + Arg(z_2)$

$Arg\left(\frac{z_1}{z_2}\right) = Arg(z_1) - Arg(z_2)$

$Re\left(\frac{1}{x+iy}\right) = \frac{x}{x^2+y^2} = \frac{Re(z)}{|z|^2}$, $Img\left(\frac{1}{x+iy}\right) = \frac{-y}{x^2+y^2} = -\frac{Im(z)}{|z|^2}$

$z^2+a^2 = z^2 - i^2a^2 = (z)^2 - (ia)^2 = (z+ia)(z-ia)$

$z_1z_2 = r_1r_2e^{i(\theta_1+\theta_2)}$, $\frac{z_1}{z_2} = \frac{r_1}{r_2}e^{i(\theta_1-\theta_2)}$

Chapter No.2 Matrices / Determinants

Square Matrix → order $n \times n$ like 2×2 or 3×3 etc (same Row + Col nos.)

Rectangular matrix → order $m \times n$ like 2×3 or 3×2 etc

↳ different No. of Rows & cols.

Scalar matrix: If $A = [a_{ij}]_{n\times n}$ then $\begin{cases} a_{ij} = 0, & \text{for } i \neq j \\ a_{ij} = K, & \text{for } i = j \end{cases}$ ↳ Non-Zero

Upper Triangular matrix, if $A = [a_{ij}]_{m\times n}$ $\Rightarrow a_{ij} = 0$ for $i > j$ → lower Elements of Diagonal are Zero

Lower Triangular matrix, $a_{ij} = 0,\ i < j$ → upper elements of diagonal are Zero

Symmetric If $A^t = A$, Skew symmetric if $A^t = -A$

Matrix Multiplication possible if $(m \times \boxed{n})(\boxed{n} \times p)$ → Equal then possible & Ans. has $m\times p$ order

$(AB)^t = B^tA^t$, $(AB)^{-1} = B^{-1}A^{-1}$

$(A+B)^t = A^t + B^t$

For any square matrix A.

$P = P + Q$ = Symmetric + Skew Symmetric matrix

$P = \frac{1}{2}(A+A^t)$ → Symmetric ; $Q = \frac{1}{2}(A-A^t)$ → Skew Symmetric

For any square matrix A, $(A^n)^t = (A^t)^n$ ✓

② **Minor of Element a_{ij}:** If $A=[a_{ij}]_{m\times n}$ then $M_{ij}=\left| \quad \right|_{(n-1)\times(n-1)}$

Cofactor of Element a_{ij}: $A_{ij}=(-1)^{i+j}M_{ij}$

$|A| = a_{11}A_{11}+a_{21}A_{21}+a_{31}A_{31}$ → Exp. from 1st col. OR $|A|=a_{11}A_{11}+a_{12}A_{12}+a_{13}A_{13}$ → Expansion from 1st Row.

Singular Matrix If $|A|=0$ otherwise Non-singular

Adjoint of Matrix A ↳ Square

$$\text{adj}\,A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^T = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

For Non-singular Matrix

$AA^{-1}=A^{-1}A=I$, $\boxed{A^{-1}=\frac{1}{|A|}\text{adj}(A)}$

Properties of Determinants i) $|A|=|A^t|$ (ii) Changing two Row/Cols results in -ve sign. $\Rightarrow |B|=-|A|$

(iii) If two Rows/col. are same $\Rightarrow |A|=0$
OR If anyone Row/col equal zero $\Rightarrow$

(iv) $|B|=K|A|=\begin{vmatrix} ka_{11} & ka_{12} & ka_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$

v) If $A=\begin{bmatrix} a_{11}+b_{11} & a_{12} & a_{13} \\ a_{21}+b_{21} & a_{22} & a_{23} \\ a_{31}+b_{31} & a_{32} & a_{33} \end{bmatrix} \Rightarrow |A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} b_{11} & a_{12} & a_{13} \\ b_{21} & a_{22} & a_{23} \\ b_{31} & a_{32} & a_{33} \end{vmatrix}$

vi) If $|A|=\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \Rightarrow |A| = \begin{vmatrix} a_{11}+ka_{21} & a_{12}+ka_{22} & a_{13}+ka_{23} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$

vii) For a triangular matrix (upper or Lower)

$|A|$ = product of diagonal elements.

$\boxed{\text{Echelon}}$ **vs Reduced Echelon Form of a matrix**

* 1st non-zero element in each Row is 1.
* Entries below leading 1 must be zero (upper triangular form)

→ **Reduced** like an Identity Matrix

Rank of matrix = No. of non-zero rows is called Rank of matrix (Row Rank)
↳ can be found by finding Echelon/Reduced Echelon form.

System of Linear Equations.

$a_1x+b_1y+c_1z=k_1$
$a_2x+b_2y+c_2z=k_2$
$a_3x+b_3y+c_3z=k_3$

i) Homogeneous → Constants (k_1,k_2,k_3) must be zero

ii) Non-Homogeneous → Constants are Non-zero. $\boxed{AX=0}$ → General form

③ Trivial Solution for Homogeneous eqn $AX=0$, $|A| \neq 0$ & $x=0$, $y=0$, $z=0$

↳ System is consistent, a unique solution exists. Always

Non-Trivial Solution for Homogeneous linear Eqns. If $|A|=0$

Then the system has Non-trivial sol → Always has Infinite many solution

↳ system is always consistent

Row Rank(A) = Rank of A_b < No of variables.

Consistent System

* Homogeneous system is always consistent

→ Unique solution
→ Trivial
→ $|A| \neq 0$

Rank(A) = Rank(A_b) = No of variable

Infinite many sol.

$|A| = 0$

Non-Trivial Sol.

Rank(A) = Rank(A_b) < # of variables

Inconsistent System

* No Solution
* Rank $A \neq$ Rank (A_b), A_b, $|A|=0 \rightarrow$ Singular.

Augmented Matrix $A_b = \begin{bmatrix} a_1 & b_1 & c_1 & : & k_1 \\ a_2 & b_2 & c_2 & : & k_2 \\ a_3 & b_3 & c_3 & : & k_3 \end{bmatrix}$ → Non-Homogeneous. System

Methods to Solve Non-Homogeneous System of Eqns. (A_b)

→ Matrix Inversion ⇒ $X = A^{-1}B = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{|A|} \text{adj}(A) \cdot \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix}$.
→ Gauss Elimination Method (Echelon form)
→ Gauss Jordan Method (Reduced Echelon form).
→ Cramer's Rule.

Vectors in Space (Chapter #3)

* Exercise 3.1 is formative

* Scalar/DOT Product $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta$, $0° \le \theta \le 180°$

$\hat{i}\cdot\hat{i} = \hat{j}\cdot\hat{j} = \hat{k}\cdot\hat{k} = 1$, $\hat{i}\cdot\hat{j} = \hat{j}\cdot\hat{k} = \hat{k}\cdot\hat{i} = 0$ If vectors are $\perp$ar, $\vec{a}\cdot\vec{b}=0$

If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$
$\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ ⇒ $\boxed{\vec{a}\cdot\vec{b} = a_1b_1 + a_2b_2 + a_3b_3}$

$\vec{a}\cdot\vec{b} = \vec{b}\cdot\vec{a}$
Commutative Law holds

④ Direction Angles & Direction Cosines.

If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \Rightarrow |\vec{r}| = \sqrt{x^2+y^2+z^2}$

Then $\cos\alpha, \cos\beta$ & $\cos\gamma$ are called Direction Cosines.

α, β, γ are called direction angles

$\Rightarrow \cos\alpha = \dfrac{x}{|\vec{r}|}, \quad \cos\beta = \dfrac{y}{|\vec{r}|}, \quad \cos\gamma = \dfrac{z}{|\vec{r}|}$

$\Rightarrow \boxed{\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1} \Rightarrow \boxed{\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2}$

$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, x, y, z are called Direction Ratios

Also, $\boxed{\hat{r} = \cos\alpha\,\hat{i} + \cos\beta\,\hat{j} + \cos\gamma\,\hat{k}}$, $\hat{r}$ is the unit vector.

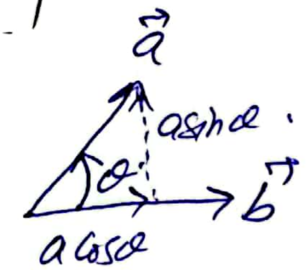
Angle b/w Two Vectors

As $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta$.

$\cos\theta = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{a}|.|\vec{b}|} \Rightarrow \boxed{\theta = \cos^{-1}(\hat{a}\cdot\hat{b})}$

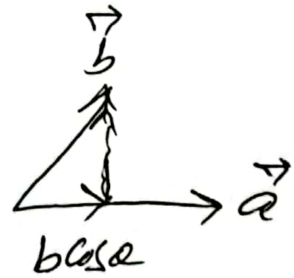
* Projection of vector $\vec{a}$ along $\vec{b}$

$\boxed{a\cos\theta = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = \vec{a}\cdot\hat{b}}$



* Projection of vector $\vec{b}$ along $\vec{a}$

$\boxed{b\cos\theta = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{a}|} = \hat{a}\cdot\vec{b}}$



Work Done by a Force

$W = \vec{F}\cdot\vec{d} = Fd\cos\theta$

If a force $\vec{F}$ moves an object from $A(x_1, y_1, z_1)$ to $B(x_2, y_2, z_2)$

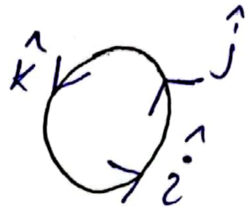
Then $\vec{d} = \vec{AB} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$

Cross Product

$\vec{a}\times\vec{b} = |\vec{a}||\vec{b}|\sin\theta\,\hat{n} \Rightarrow \vec{a}\times\vec{b} \perp \vec{a} \Rightarrow (\vec{a}\times\vec{b})\cdot\vec{a} = 0$

$\hat{n}$ is normal to the plane having $\vec{a}$ & $\vec{b}$. Also, $\vec{a}\times\vec{b} \perp \vec{b} \Rightarrow (\vec{a}\times\vec{b})\cdot\vec{b} = 0$

(5) $\vec{a}\times\vec{b} = -\vec{b}\times\vec{a}$ (Anti-Commutative)



$\hat{i}\times\hat{j} = \hat{k}, \quad \hat{j}\times\hat{k} = \hat{i}, \quad \hat{k}\times\hat{i} = \hat{j}$

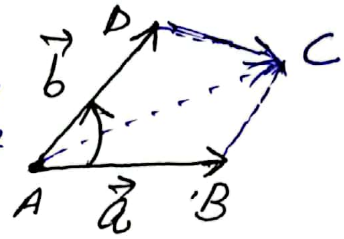
$\Rightarrow \hat{j}\times\hat{i} = -\hat{k}, \quad \hat{k}\times\hat{j} = -\hat{i}, \quad \hat{i}\times\hat{k} = -\hat{j}$

$\hat{i}\times\hat{i} = \hat{j}\times\hat{j} = \hat{k}\times\hat{k} = 0$ As $\sin 0^\circ = 0$

If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ & $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

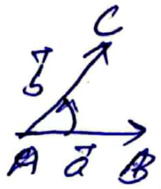
$$\Rightarrow \vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

* $\vec{a}$ & $\vec{b}$ are Two Adjacent sides



(*) $|\vec{a}\times\vec{b}|$ = Area of parallelogram ABCD

Area of Triangle ABC $= \frac{1}{2}|\vec{a}\times\vec{b}|$



* If points are given like $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$, $C(x_3, y_3, z_3)$

then $\vec{a} = \vec{AB} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$

$\vec{b} = \vec{AC} = (x_3 - x_1)\hat{i} + (y_3 - y_1)\hat{j} + (z_3 - z_1)\hat{k}$

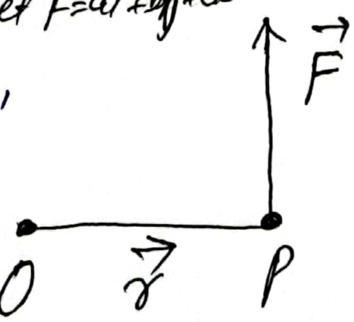
* If $\vec{a}\times\vec{b} = \vec{0}$, the vectors are either parallel or anti parallel.

<u>Angle b/w Two vectors</u> As $\vec{a}\times\vec{b} = |\vec{a}||\vec{b}|\sin\theta\,\hat{n}$

$$\Rightarrow \theta = \sin^{-1}\left(\frac{|\vec{a}\times\vec{b}|}{|\vec{a}||\vec{b}|}\right) \quad 0 \le \theta \le \pi$$

<u>Moment of Force or Torque</u>

Let $F = a\hat{i} + b\hat{j} + c\hat{k}$



* If force $\vec{F}$ applied on point 'P'

* <u>Finding Moment of Force/Torque about point O</u> then

Moment Arm $= \vec{r} = \vec{OP} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\Rightarrow \vec{\tau} = \vec{r}\times\vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ a & b & c \end{vmatrix}$$

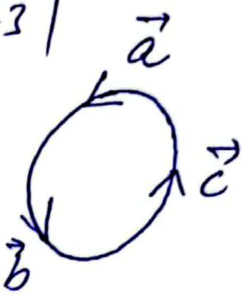
⑥ Scalar Triple Product $\vec{a}\cdot(\vec{b}\times\vec{c})$ or $[\vec{a}\ \vec{b}\ \vec{c}]$

If $\vec{a} = a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$, $\vec{b} = b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$, $\vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$

$$\vec{a}\cdot(\vec{b}\times\vec{c}) = [\vec{a}\ \vec{b}\ \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

* $\vec{a}\cdot(\vec{b}\times\vec{c}) = (\vec{a}\times\vec{b})\cdot\vec{c}$ ✓

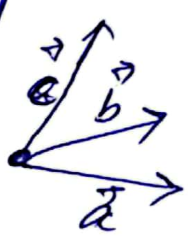
* $\vec{a}\cdot(\vec{b}\times\vec{c}) = \vec{b}\cdot(\vec{c}\times\vec{a}) = \vec{c}\cdot(\vec{a}\times\vec{b})$



* $\vec{a}\cdot\vec{b}\times\vec{c} = -\vec{b}\cdot\vec{a}\times\vec{c} = \vec{a}\cdot\vec{c}\times\vec{b}$

$\Rightarrow \hat{i}\cdot\hat{j}\times\hat{k} = \hat{i}\cdot\hat{i} = 1 \Rightarrow \hat{j}\cdot\hat{k}\times\hat{i} = 1$, $\hat{k}\cdot\hat{i}\times\hat{j} = 1$

Also, Volume of parallelopiped $= |\vec{a}\cdot(\vec{b}\times\vec{c})|$ where $\vec{a},\vec{b},\vec{c}$ represent adjacent / coterminous sides having same (vertex)



Volume of Tetrahedron

$$= \frac{1}{6}(\vec{a}\cdot\vec{b}\times\vec{c})$$

* Find Volume of Tetrahedron / parallelopiped if vertices are given. A, B, C, D.

Then $\vec{a} = \overrightarrow{AB}$, $\vec{b} = \overrightarrow{AC}$, $\vec{c} = \overrightarrow{AD}$.

Then find $|\vec{a}\cdot\vec{b}\times\vec{c}|$ for parallelopiped volume

and $\frac{1}{6}|\vec{a}\cdot\vec{b}\times\vec{c}|$ for Tetrahedron Volume.

Coplanar Vectors

If two or more vectors lie in the same plane, we call them coplanar vectors.

$\Rightarrow \vec{a}\cdot\vec{b}\times\vec{c} = 0 \Rightarrow \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$ Always for coplanar vectors

* If two vectors are parallel, $\vec{a} = a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$, $\vec{b} = b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$

$\Rightarrow \frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = k \longrightarrow$ Their direction Ratios are equal.

⑦

Chap # 04 Sequence & Series

* a_n is called General term or nth term
  e.g. $a_n = (-1)^{n+1}(3n-5) \rightarrow$ To find 1st 4 terms put $n = 1, 2, 3, 4$ respectively.

* **Arithmetic Sequence/progression (A.P)**

* General Term: $\longrightarrow$ $\boxed{a_n = a_1 + (n-1)d} \Rightarrow$ called Recursive Formula
  $a, a+d, a+2d, a+3d, \ldots \ldots$

  * $a_1 =$ 1st term, $d =$ Common Difference
    $d = a_n - a_{n-1}$

  e.g. $d = a_2 - a_1 = a_3 - a_2 = a_4 - a_3$ etc.

Example: $a_5 = a_1 + 4d$, $a_{19} = a_1 + 18d$ etc.

* **Arithmatic Mean b/w Two Numbers**

$A = A.M = \frac{a+b}{2}$ if a, A, b are in A.P.
i.e. $A - a = b - A = d$.

Example:- Find 4 A.Ms' b/w 19 & 54.
$19, A_1, A_2, A_3, A_4, 54$, here $n = 6$, $\boxed{a_1 = 19}$ $\boxed{a_6 = 54}$

As $a_6 = a_1 + 5d \Rightarrow 54 = 19 + 5d \Rightarrow \boxed{d = 7}$, $A_1 = a_1 + d$, $A_2 = a_1 + 2d$ etc.

* **Arithmetic Series** $\Rightarrow$ Sum $\Rightarrow S_n = a_1 + a_2 + \cdots + a_n$

When a_n is given: $S_n = \frac{n}{2}[a_1 + a_n]$ OR $S_n = \frac{n}{2}[2a_1 + (n-1)d]$ when a_n is not known.

To find total terms from A to B successive integers.
$n = B - A + 1$. e.g. $23, 24, 25, \ldots \ldots, 75$ $n = ??$
$n = 75 - 23 + 1 \Rightarrow \boxed{n = 53 \text{ terms}}$

* **Important** For any AP series like $1+2+3+4+\cdots+a_n$
S_n gives you sum of 1st n terms. Example, If $n = 1$ the $S_1 = 1$, for $n = 2$, $S_2 = a_1 + a_2 = 1 + 2 = 3$, for $n = 3$ $S_3 = 1 + 2 + 3 = 6$ etc.

8

How to Identify A.P in Word Problems:-

* Look for a value that increases or decreases by the same amount etc.
* Added each year/month/day etc. * Tree height grows 5cm every week.
(A.*) Constant rate of change (very important) in A.P.
* Growth type is Linear or Addition
* Saves Rs. 'x' more than previous year etc.
* Salary increase

Geometric Sequence

* Seats increased by — in each row.

* Identification in word problems:-

r = Common Ratio.

$$a, ar, ar^2, \ldots, ar^{n-1} \longrightarrow a_n$$

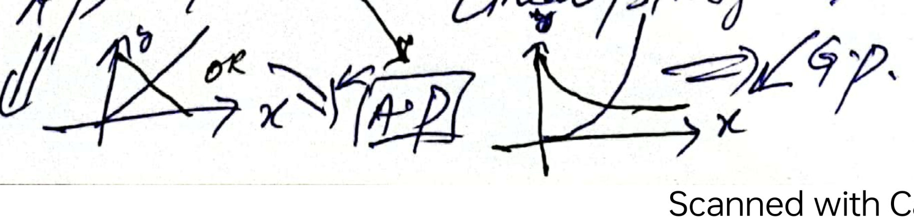
$$r = \frac{a_2}{a_1} = \frac{a_3}{a_2} = \frac{a_n}{a_{n-1}} \text{ etc}$$

Key points for identification

* Grows/decay by a fixed Percentage / Times, 2 times etc.
  eg. * population grows by 5% yearly.
  * Car value decreases by 10% annually.
  * investment increases by 8% each year etc.
* If each quantity is * doubled, tripled, halved or multiplied by a fixed factor, then it is a G.P problem. * Reduced to one-third etc.

More Examples:- 80% of the previous, doubles every hour, depreciates by 15%, grows exponentially, * Constant Ratio $\frac{ar}{a} = r$, $\frac{ar^2}{ar} = r$ etc.

In G.P, we have Exponential Curve (due to multiplication) while in A.P, Linear/straight line graph



9 More on G.P. General Term / nth term

$a_n = ar^{n-1}$, You can any term by putting 'n' value

$a_2 = ar$, $a_3 = ar^2$, $a_{10} = ar^9$ etc

Geometric Mean (G.M.) $G = \pm\sqrt{ab}$ if a, G, b in G.P.

- +ve if a & b are +ve
- −ve if both a & b are −ve
- imaginary if both have opposite signs

$\frac{G}{a} = \frac{b}{G} \Rightarrow G^2 = ab$

Ex Finding two GM's b/w 81 & 3 -

$81, G_1, G_2, 3$, $\boxed{a_1 = 81}$ $a_4 = 3$, $ar^3 = 3 \Rightarrow 81r^3 = 3$.

$\Rightarrow r^3 = \frac{1}{27} \Rightarrow \boxed{r = \frac{1}{3}}$ Thus, $G_1 = ar$, $G_2 = ar^2$

*** Sum of n terms of G.P** $\boxed{S_n = \frac{a_1(1-r^n)}{1-r}}$, $r \neq 1$, $|r| < 1$, $-1 < r < 1$

And $\boxed{S_n = \frac{a_1(r^n - 1)}{r-1}}$ if $|r| > 1$

$r < -1$ & $r > 1$.

*** Special case** when 'n' is not known. You can 1st find 'n' by $\boxed{a_n = a_1 r^{n-1}}$ OK.

directly use $\boxed{S_n = \frac{a_1 - a_n r}{1-r}}$

*** Infinite Geometric Series.** $\boxed{S_\infty = \frac{a_1}{1-r}}$ But $|r| < 1$, $-1 < r < 1$

*** Harmonic Sequence (H.P)**

If the reciprocal of the terms are in A.P, then it is called H.P. eg $1, \frac{1}{4}, \frac{1}{7}, \frac{1}{10}, \ldots$ are in H.P

because $1, 4, 7, 10, \ldots$ are in A.P.

General term: $\frac{1}{a_1}, \frac{1}{a_1+d}, \frac{1}{a_1+2d}, \ldots$ in H.P.

(10) The general term of the H.P is

$$\boxed{\frac{1}{a_n} = \frac{1}{a_1+(n-1)d}} \xrightarrow[\text{Reciprocal}]{\text{Take}} \boxed{a_n = a_1+(n-1)d} \text{ in A.P.}$$

* **Harmonic Mean** If a, H, b in H.P then $\frac{1}{a}, \frac{1}{H}, \frac{1}{b}$ are in A.P. $\Rightarrow \frac{1}{H} - \frac{1}{a} = \frac{1}{b} - \frac{1}{H}$

By simplifying, $\boxed{H = \frac{2ab}{a+b}}$

(7A) If a & b are +ve i.e. $a>0$ & $b>0$

then $\boxed{A > G > H}$

If a and b are −ve i.e. $a<0$ & $b<0$

then $\boxed{A < G < H}$

and in both cases, $\boxed{G^2 = AH}$

Arithmetico-Geometric Series (* has both A.P & G.P in it)

e.g $\frac{0}{1}, \frac{1}{2}, \frac{2}{4}, \frac{3}{8}, \frac{4}{16}, \ldots$ *

* **General Term** $\boxed{T_n = A_n G_n = [a+(n-1)d]\, r^{n-1}}$

* **Sum of A.G.P.**

$$\boxed{S_n = \frac{a}{1-r} + \frac{dr\,(1-r^n)}{(1-r)^2} - \frac{(a+nd)r^n}{1-r}}$$

a = 1st term of A.P

d = Common Difference of A.P.

r = Common ratio of G.P. $\boxed{S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}}$ $|r|<1$

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Chapter # 05

$\boxed{\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}}$

If you have a polynomial $P(x) = - - - -$ and is divided by $x-a$, To find the remainder put $x - a = 0 \Rightarrow \boxed{x = a}$ Now find $\boxed{P(a) = \text{Remainder}}$

* A number 'a' is a Zero of a $P(x)$ if $P(a) = 0$

* Factor Theorem If "a" is a zero of a $P(x)$ then $x = a \Rightarrow (x-a)$ is a factor of $P(x)$

* Cubic polynomial:

Let $P(x) = ax^3 + bx^2 + cx + d$

Rational Roots Let $p = d$ and $q = a$

* Always take +ve values of a & d.

Let $\boxed{p = 6}$

Roots/Zeros of p: $\pm 1, \pm 2, \pm 3, \pm 6$.

Roots/zeros of q: $\pm 1, \pm 2$ then.

$\boxed{\text{Let } 2}$

Possible Rational Roots are: $\frac{p}{q} = \pm 1, \pm 2, \pm 3, \pm 6, \pm \frac{3}{2}$

* Must use Calculator to find Roots of cubic polynomial

CH#6 Permutation and Combination (1) * Fundamental principle of Counting

$n! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1 \Rightarrow n! = n(n-1)!$

if $n \cdot (n-1) \Rightarrow \frac{n(n-1)(n-2)!}{(n-2)!} \Rightarrow \frac{n!}{(n-2)!}$

(12) If order/arrangement matters then it is permutation otherwise it is a combination problem.

Permutation	Combination
Think of position/arrangement	* Think of selection/choosing.
* Check if it matter if you swap two things	* selecting objects.
* Seating arrangements	* Forming committees
* passwords.	* Choosing Teams.
* Rankings * Race positions	* Making Groups.
	* pick objects

$\boxed{{}^nP_r = r! \times {}^nC_r}$

* n must be distinct

$${}^nP_r = \frac{n!}{(n-r)!}, \quad {}^nC_r = \frac{n!}{r!(n-r)!}$$

$\Rightarrow {}^nC_r = \frac{{}^nP_r}{r!} \Rightarrow$

* **Permutations of Repeated Objects.**

No. of ways/arrangements/permutations

$$= \frac{n!}{n_1! \times n_2! \cdots \times n_k!}$$

e.g Evening ($n=7$) $\Rightarrow$ No. of ways $= \frac{7!}{2!\cdot 2!}$

E → 2 times
n → 2 times

Circular Permutation

For Living things No. of ways $= (n-1)!$ → can't be flipped. or letters like ABCD or numbers 1234 etc.

For Non-living things " $= \frac{(n-1)!}{2} \Rightarrow$ can be flipped beads in necklace. flowers in Garland

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Important Results

$$^nP_0 = 1, \quad ^nP_1 = n, \quad ^nP_n = n!$$

$$^nC_0 = 1, \quad ^nC_1 = n, \quad ^nC_n = 1.$$

$$^nC_r = {}^nC_{n-r} \qquad \binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$$

Chapter # 07

For mathematical Induction $\rightarrow$ 2 steps.
i) Step 1 (Basis Step)
put $n = 1$.

Step 2 (inductive Step).
Assume $P(k)$ is true so, $P(k+1)$ must be true.
$P(n)$ is called proposition.
* Use plugin method for MCQs

Binomial Theorem $(a+b)^n = \binom{n}{0}a^n b^0 + \binom{n}{1}a^{n-1}b^1 + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{n-1}a^1 b^{n-1} + \binom{n}{n}b^n$

'n' must be +ve integer.

If n is power, total terms are $n+1$

Middle term if n is Even. $\left(\frac{n+2}{2}\right)^{th}$ term

Middle term if n is odd $\left(\frac{n+1}{2}\right)^{th}$ & $\left(\frac{n+3}{2}\right)^{th}$ terms

General Term :- $\boxed{T_{r+1} = \binom{n}{r} a^{n-r} \cdot b^r}$

A term 'r'th position from the end in the expansion of $(a+b)^n$ is at $(n+2-r)^{th}$ position from beginning

(14) To find constant term (term independent of x) means $x^0 \rightarrow$ Use general term formula.

* **Binomial Series** when power is -ve / fraction

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \cdots \infty \text{ terms.}$$

$$T_{r+1} = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}x^r$$

The Binomial Series converges only if

$$|r|<1 \Rightarrow |x|<1 \Rightarrow -1<x<1$$

To find r Divide 2nd term by 1st term

eg When this series is convergent

$$\left(3+\frac{2}{x}\right)^{-\frac{1}{3}}, \text{ take } \left|\frac{2}{x} \div 3\right| < 1.$$

$$\Rightarrow \left|\frac{2}{x} \times \frac{1}{3}\right| < 1 \Rightarrow \left|\frac{2}{3x}\right| < 1$$

$$\Rightarrow |2| < |3x| \Rightarrow |3x| > 2$$

$$\Rightarrow \boxed{|x| > \frac{2}{3}} \text{ OR } x < -\frac{2}{3} \text{ or } x > \frac{2}{3}$$

$$\pm x > \frac{2}{3}, \; x > \frac{2}{3} \text{ or } -x > \frac{2}{3} \Rightarrow x < -\frac{2}{3}$$

(15) For Unit Digit, must Divide by 10 to get the Remainder

$17^{203} = 7^{203}$

17 → unit digit of Base

* Check the length of repeat cycle of 7 $7^5 = 16807$

$7^1 = 7$, $7^2 = 49$, $7^3 = 343$, $7^4 = 2401$

So, $\boxed{\text{Cycle length} = 4}$ ⇒ $\boxed{\overset{7^1}{7}, \overset{7^2}{9}, \overset{7^3}{3}, \overset{7^4}{1}}$

Next, divide Exponent/power by cycle length and check the remainder $\frac{203}{4} = \frac{4 \times 50 + 3}{4} = 3.$

$4 \overline{)203}$ = 50, 200, remainder 3

$\boxed{\text{Remainder} = 3}$ So

$7^{203} = 7^3 = 34\boxed{3}$ ✓ 17 for

divide $10 \overline{)343}$ = 34, 340, remainder 3

OR. $7^3 =$ 3 → from cycle

Chapter # 8

* Distance formula b/w two Points $P(x_1, y_1)$, $Q(x_2, y_2)$

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

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$$\boxed{\begin{aligned} \cos(\alpha-\beta) &= \cos\alpha\cos\beta + \sin\alpha\sin\beta, \quad \alpha > \beta \\ \cos(\alpha+\beta) &= \cos\alpha\cos\beta - \sin\alpha\sin\beta \\ \sin(\alpha-\beta) &= \sin\alpha\cos\beta - \cos\alpha\sin\beta \\ \sin(\alpha+\beta) &= \sin\alpha\cos\beta + \cos\alpha\sin\beta \end{aligned}}$$

S | A
T | C

Allied Angle Formulas

odd

sign

If n is odd

$\sin\left(n \times \frac{\pi}{2} \pm \alpha\right) = \pm \cos\alpha$

$\sin \leftrightarrow \cos$
$\operatorname{cosec} \leftrightarrow \sec$
$\tan \leftrightarrow \cot$

↳ check the quadrant and see if sin is +ve or −ve in this Quadrant and use that sign befor $\cos\alpha$.

* If n is even then $\sin \leftrightarrow \sin$
$\cos \leftrightarrow \cos$
etc.

$$\boxed{\tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}}$$

$$\boxed{\tan(\alpha-\beta) = \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}}$$

$$\cos^2\alpha + \sin^2\alpha = 1$$

* $a\sin\theta \pm b\cos\theta = r\sin(\theta \pm \phi)$

$a = r\cos\phi, \ \pm b = r\sin\phi$

$r = \sqrt{a^2+b^2}, \quad \phi = \tan^{-1}(b/a)$

(17)

* $\sin 2\alpha = 2\sin\alpha\cos\alpha \Rightarrow \sin\alpha = 2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}$

$\cos 2\alpha = \cos^2\alpha - \sin^2\alpha = 2\cos^2\alpha - 1 = 1 - 2\sin^2\alpha$

$$\boxed{\cos^2\alpha = \frac{1+\cos 2\alpha}{2}} \Rightarrow 2\cos^2\alpha = 1+\cos 2\alpha.$$

$$\cos\alpha = \pm\sqrt{\frac{1+\cos 2\alpha}{2}}$$

$$\cos^2\frac{\alpha}{2} = \frac{1+\cos\alpha}{2} \Rightarrow \cos\frac{\alpha}{2} = \pm\sqrt{\frac{1+\cos\alpha}{2}}$$

$$\boxed{\sin^2\alpha = \frac{1-\cos 2\alpha}{2}} \Rightarrow \sin\alpha = \pm\sqrt{\frac{1-\cos 2\alpha}{2}}$$

$$\sin^2\frac{\alpha}{2} = \frac{1-\cos\alpha}{2} \Rightarrow \sin\frac{\alpha}{2} = \pm\sqrt{\frac{1-\cos\alpha}{2}}$$

$$\boxed{\tan^2\alpha = \frac{1-\cos 2\alpha}{1+\cos 2\alpha}} \qquad \boxed{\tan 2\alpha = \frac{2\tan\alpha}{1-\tan^2\alpha}}$$

$$\boxed{\sin(3\alpha) = 3\sin\alpha - 4\sin^3\alpha} \qquad \boxed{\cos 3\alpha = 4\cos^3\alpha - 3\cos\alpha}$$

$$\boxed{\tan 3\alpha = \frac{3\tan\alpha - \tan^3\alpha}{1-3\tan^2\alpha}.}$$

$$\boxed{\begin{aligned} 2\sin\alpha\cos\beta &= \sin(\alpha+\beta) + \sin(\alpha-\beta) \quad \alpha > \beta \\ 2\cos\alpha\sin\beta &= \sin(\alpha+\beta) - \sin(\alpha-\beta) \\ 2\cos\alpha\cos\beta &= \cos(\alpha+\beta) + \cos(\alpha-\beta) \\ -2\sin\alpha\sin\beta &= \cos(\alpha+\beta) - \cos(\alpha-\beta) \end{aligned}}$$

$$\begin{aligned} \sin\alpha + \sin\beta &= 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) \\ \sin\alpha - \sin\beta &= 2\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right) \\ \cos\alpha + \cos\beta &= 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) \\ \cos\alpha - \cos\beta &= -2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right) \end{aligned}$$

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Chapter # 09

$[a, b] \to$ is called closed interval
$a \le x \le b$

$(a, b) \to$ open interval $a < x < b$

Domain and Range of $\sin x$ and $\cos x$

$D_{\sin} = (-\infty, \infty)$ or $\mathbb{R} \longrightarrow D_{\cos}$ ← Always

$R_{\sin} = R_{\cos x} = [-1, 1] \quad -1 \le y \le 1$

$-1 \le \sin\theta, \cos\theta \le 1$

If $-2 \le 2\sin\theta \le 2$ etc

$\boxed{\sec\theta = \frac{1}{\cos\theta}}$

$\tan\theta = \frac{\sin\theta}{\cos\theta}$, $\cos\theta \ne 0 \Rightarrow \theta \ne \cos^{-1}(0)$
$\Rightarrow \theta \ne (2n+1)\frac{\pi}{2}$
↳ odd multiples

$D_{\tan\theta} = \mathbb{R} - (2n+1)\frac{\pi}{2}$

$\operatorname{cosec}\theta = \frac{1}{\sin\theta}$

$\cot\theta = \frac{\cos\theta}{\sin\theta}$, $\sin\theta \ne 0 \Rightarrow \theta \ne \sin^{-1}(0) \ne n\pi$
↓ all integers

$\text{Range}_{\tan, \cot} \Rightarrow (-\infty, \infty) = \mathbb{R}$

Range of $\sec\theta$, $\operatorname{cosec}\theta$, $|y| \ge 1$
$\Rightarrow y \ge 1, y \le -1$

Period of Trig. fn

$A\cos(Bx + C) + D$.
↳ sin, sec, cosec

$$\boxed{T = \frac{2\pi}{|B|}}$$

$A\tan(Bx + C) + D$ or cot

$$\boxed{T = \frac{\pi}{|B|}}$$

(19) e.g. period of $\tan\frac{5x}{7} = \frac{\pi}{\frac{5}{7}} = \frac{7\pi}{5}$

* period is always +ve.

Example Find the period of $f(x) = \cot 3x + \sin\frac{2x}{3}$.

period of $\cot 3x = \frac{\pi}{3}$ | period of $\sin\frac{2x}{3} = \frac{2\pi}{\frac{2}{3}} = \frac{3\pi}{1}$

Period of $f(x) = \frac{\text{LCM of Numerators}}{\text{HCF of Denominator}} = \frac{LCM(\pi, 3\pi)}{HCF(3,1)} = 3\pi$

Max. and Min Value of a function

$y = A\cos(Bx + C) + D$

C → Horizontal translation ; D → Vertical translation

Max value :- $D + |A|$, Min value $= D - |A|$

Amplitude is always +ve $= |A|$

e.g. Max (M) and Min (m) value of $\frac{1}{3} - 5\cos x$

$M = \frac{1}{3} + |-5| = 5 + \frac{1}{3} = \frac{16}{3}$, $m = \frac{1}{3} - |-5| = \frac{1}{3} - 5 = -\frac{14}{3}$.

Rule When both M and m are +ve then
Max $= M' = m$ and Min $= m' = M$.
just swap both

Even & odd fn. If $f(-x) = -f(x) \rightarrow$ odd → sin, tan
$f(-x) = f(x) \rightarrow$ Even → Cos, Sec

If $f(x) \neq \pm f(x) \rightarrow$ Neither Even nor odd like $\sin x + \cos$

Odd → symmetric about origin

Even fn → Symmetric w.r.t. y-axis

$|A| = \frac{\text{Max} - \text{Min}}{2}$, $D = \frac{\text{Max} + \text{Min}}{2}$

Radius of Ferris Wheel = Amplitude.

If starts at lowest pa then $y = -A\cos(Bt) + D$.